

RITESH PATIL

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Summary

Data Management Engineer I with experience in **Generative AI**, **Multi-Agent Systems**, and enterprise AI solutions. Skilled in developing AI chatbots using **MCP tools**, **RAG**, **PostgreSQL (pgvector)**, **Neo4j**, and **Python**, with a focus on building scalable, intelligent applications for enterprise environments.

Experience

Data Management Engineer I (AI & Data) – Deloitte USI

Feb 2026 – Present

- Developed an enterprise AI chatbot for a leading insurance client using a multi-agent architecture and MCP tools to enable intelligent query handling.
- Implemented RAG using PostgreSQL (pgvector) and Neo4j for semantic search and graph-based knowledge retrieval, delivering scalable enterprise AI solutions.

AI Engineer Intern – OnPoint Software Services

Jul 2025 – Dec 2025

- Developed real-time AI voice and chat assistants using LangGraph, Google Gemini, LiveKit, and pgvector.
- Built scalable backend services with FastAPI and Docker, enabling efficient deployment and concurrent request handling.

Skill Set

Programming Languages: Python, SQL

Generative AI & LLMs: LangChain, LangGraph, RAG, Pydantic, MCP

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, Keras

Cloud: AWS

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Power BI, Excel

Development & Deployment: FastAPI, Flask, Django, Streamlit, Docker, Git, MLflow, DVC

Databases & Vector Stores: MySQL, PostgreSQL, PGVector, MongoDB, Neo4J

Tools & Platforms: LangSmith, Postman, Jupyter Notebooks, N8N

Projects

Vital Voice – AI Voice Gym & Dietician Coach

Created a **real-time low latency AI voice assistant** offering gym and diet coaching through natural voice conversations. Designed a multi-agent architecture with dedicated trainer and dietician agents to provide personalized fitness guidance.

Tech: FastAPI, LiveKit, Google Gemini, LangGraph, PostgreSQL

Scribble.io – Air Canvas Math Solver

Built an AI-powered system enabling users to draw math equations in the air using hand gestures. Integrated OpenCV and Gemini for real-time equation recognition and automated solution generation.

Tech: Django, OpenCV, Cvzone, Google Gemini

AI-Based Advanced Proctoring System

Designed an intelligent proctoring solution for online exams using YOLO and R-CNN models to detect suspicious activities and maintain test integrity. Enhanced monitoring through real-time object and behaviour detection.

Tech: Deep Learning, Object Detection, YOLO, R-CNN

Certification

- [**AWS Cloud Practitioner**](#)
- [**Oracle Cloud Infrastructure 2024 Generative AI Certified Professional**](#)
- [**Claude Certified Developer**](#)
- [**Machine Learning Fundamentals Micro-Credential Issued by Alteryx**](#)

Academic

Bharati Vidyapeeth College of Engineering | Bachelor of Technology
Major in Computer Science& Engineering; Specialization in **AI-ML**.

(2021-2025)
CGPA 8.57/10